

# Species-Aware Module Review: Aerobic Phenylacetate Catabolism in *Pseudomonas putida* KT2440

**Target taxon:** *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Module:** Aerobic phenylacetyl-CoA catabolic pathway (generic module `amino_acid_metabolism` / local bucket `kegg:ppu00360`) **Curation intent:** Manual module satisfiability + gene-annotation review (not a generic pathway essay)

## 1. Executive Summary

The aerobic phenylacetyl-CoA (paa) catabolic module is **fully satisfiable** in *P. putida* KT2440. All seven core reaction steps are encoded by a single, non-redundant, co-localized gene cluster spanning **PP\_3270–PP\_3284** (paaZ, paaABCDE, paaK, paaJ, paaI, paaH, paaG, paaF). The architecture is gene-for-gene homologous to the canonical *E. coli* W `paaZ / paaABCDEFGHIJK` catabolon (Ferrández et al. 1998,

**P** 9748275

and the activation step plus operon regulation are **directly validated in KT2440 itself** (Kim et al. 2009,

**P** 19406896

Two curation actions dominate this review:

1. **Add the missing thiolase.** Module step 7 (3-oxoadipyl-CoA thiolysis) has **no candidate** in the local metadata, yet the covering gene clearly exists: **PP\_3280 = paaJ, EC 2.3.1.174**, sitting inside the cluster between paaK and paaI. Mark step 7 **covered**, not a gap.
2. **Prune over-propagated candidates.** Only 13 of 28 candidate genes belong to the core chain. The other ~15 are peripheral or over-propagated from the umbrella KEGG *Phenylalanine metabolism* map (ppu00360) and neighboring beta-oxidation / aromatic amino-acid maps.

No lineage-specific alternatives or new GO terms are required; the pathway is the textbook version. The main caveat is that all KT2440 paa proteins are **UniProt TrEMBL (unreviewed)** — EC/subunit calls are automatic annotations anchored by synteny and *E. coli* homology.

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## 2. Target-Organism Pathway Definition

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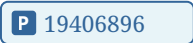
**What is included (pathway boundary):** activation of free phenylacetate to phenylacetyl-CoA, followed by *aerobic, CoA-dependent, epoxide/oxepin ring cleavage* and a beta-oxidation-like processing of the ring-opened CoA ester down to acetyl-CoA + succinyl-CoA (TCA entry). This is a **non-oxygenolytic-of-the-free-ring** strategy: the ring is epoxidized *while thioesterified to CoA*, distinguishing it from classical ring-hydroxylating dioxygenase routes.

**Neighboring processes to keep OUT of this module:** - **Phenylalanine → phenylacetate feeder chemistry** (phhA Phe-4-hydroxylase; aromatic aminotransferases; phenylacetaldehyde dehydrogenase peaE). These *supply the substrate* but are not the CoA pathway. - **Tyrosine/homogentisate catabolism** (hpd, PP\_3433). Separate KEGG map ppu00130. - **β-Ketoadipate / protocatechuate pathway** (pcaF, PP\_1377) — shares EC 2.3.1.174 with paaJ but is a different pathway. - **Generic fatty-acid β-oxidation** (fadA, PP\_2217, PP\_3726, hbd/PP\_3755) — crotonase/thiolase-superfamily paralogs.

**Alternate names / DB definitions:** "phenylacetic acid (PA) degradation", "phenylacetyl-CoA catabolon", "PA-CoA pathway". KEGG module **M00878** (phenylacetate degradation, PA-CoA pathway) is the correct module-level analogue; the local bucket points instead at the broad map **ppu00360 (Phenylalanine metabolism)**, which is why so many peripheral genes were pulled in. **Recommendation:** retarget the local bucket from ppu00360 to the paa-specific module (KEGG M00878 / MetaCyc PWY-1341 "phenylacetate degradation I aerobic").

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### 3. Expected Step Model → Coverage in KT2440

| #  | Module step                          | Enzyme / EC                                                 | KT2440 gene                                       | Status                                                                                                                       |
|----|--------------------------------------|-------------------------------------------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| 1  | Phenylacetate activation             | phenylacetate-CoA ligase, EC 6.2.1.30                       | <b>paaK / PP_3279</b>                             | covered (direct in-strain activity,<br> ) |
| 2  | Ring 1,2-epoxidation                 | phenylacetyl-CoA epoxidase (multicomponent), EC 1.14.13.149 | <b>paaA/B/C/D/E / PP_3278/3277/3276/3275/3274</b> | covered                                                                                                                      |
| 3  | Epoxide isomerization                | 1,2-epoxyphenylacetyl-CoA isomerase, EC 5.3.3.18            | <b>paaG / PP_3283</b>                             | covered                                                                                                                      |
| 4a | Oxepin-CoA hydrolysis (ring opening) | oxepin-CoA hydrolase, EC 3.3.2.12                           | <b>paaZ / PP_3270</b> (hydrolase domain)          | covered                                                                                                                      |
| 4b | Semialdehyde oxidation               | aldehyde dehydrogenase, EC 1.2.1.91                         | <b>paaZ / PP_3270</b> (ALDH domain, bifunctional) | covered                                                                                                                      |
| 5  | Open-chain enoyl-CoA hydration       | enoyl-CoA hydratase, EC 4.2.1.17                            | <b>paaF / PP_3284</b>                             | covered                                                                                                                      |
| 6  | 3-hydroxyadipyl-CoA oxidation        | 3-hydroxyacyl-CoA dehydrogenase, EC 1.1.1.35                | <b>paaH / PP_3282</b>                             | covered                                                                                                                      |
| 7  | 3-oxoadipyl-CoA thiolysis            | 3-oxoadipyl-CoA thiolase, EC 2.3.1.174                      | <b>paaJ / PP_3280</b>                             | covered — but MISSING from candidate metadata (gap in bucket, not in genome)                                                 |

**Accessory (in cluster, not a required step):** paaI / PP\_3281 (thioesterase, EC 3.1.2.-) — proofreading/editing thioesterase that hydrolyzes off-pathway CoA intermediates; keep as cluster context, mark **covered/accessory**, not a core step.

Every required step is present. **Module satisfiability = SATISFIED.**

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## 4. Candidate Genes and Evidence

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4A. Core cluster genes (high confidence — all TrEMBL, synteny + *E. coli* homology)

| Gene | Locus   | UniProt | Role                                                                           | Evidence type                                                                           | Curation note                                                                                                                                                         |
|------|---------|---------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| paaK | PP_3279 | Q88HS4  | PA-CoA ligase (EC 6.2.1.30)                                                    | <b>Direct KT2440</b><br>enzyme activity<br>( <a href="#">P 19406896</a> )<br>+ homology | Single copy; no auxiliary ligase. Nomenclature caveat: ligase = <i>paaF</i> in <i>Pseudomonas</i> sp. Y2<br>( <a href="#">P 18544072</a> )<br>— map by EC, not letter |
| paaA | PP_3278 | Q88HS5  | epoxidase $\alpha$ (EC 1.14.13.149)                                            | Homology + synteny                                                                      | Catalytic diiron oxygenase subunit                                                                                                                                    |
| paaB | PP_3277 | Q88HS6  | epoxidase regulatory/structural                                                | Homology + synteny                                                                      | 93 aa small subunit                                                                                                                                                   |
| paaC | PP_3276 | Q88HS7  | epoxidase $\beta$ (EC 1.14.13.149)                                             | Homology + synteny                                                                      |                                                                                                                                                                       |
| paaD | PP_3275 | Q88HS8  | epoxidase component (Fe-S)                                                     | Homology + synteny                                                                      | No EC in UniProt; "subunit of phenylacetate degradation enzyme"                                                                                                       |
| paaE | PP_3274 | Q88HS9  | epoxidase reductase (EC 1.14.13.149)                                           | Homology + synteny                                                                      | NAD(P)H:ferredoxin-type reductase                                                                                                                                     |
| paaG | PP_3283 | Q88HS0  | epoxide $\rightarrow$ oxepin isomerase (EC 5.3.3.18)                           | Homology + synteny                                                                      | Crotonase superfamily                                                                                                                                                 |
| paaZ | PP_3270 | Q88HT3  | <b>bifunctional</b><br>oxepin-CoA hydrolase (EC 3.3.2.12) + ALDH (EC 1.2.1.91) | Homology + synteny                                                                      | 684 aa two-domain; covers steps 4a & 4b                                                                                                                               |
| paaF | PP_3284 | Q88HR9  | enoyl-CoA hydratase (EC 4.2.1.17)                                              | Homology + synteny                                                                      | Cluster copy; paralogs elsewhere (see 5)                                                                                                                              |
| paaH | PP_3282 | Q88HS1  |                                                                                |                                                                                         | 505 aa                                                                                                                                                                |

| Gene | Locus   | UniProt | Role                                    | Evidence type      | Curation note                  |
|------|---------|---------|-----------------------------------------|--------------------|--------------------------------|
|      |         |         | 3-hydroxyacyl-CoA DH (EC 1.1.1.35)      | Homology + synteny |                                |
| paaJ | PP_3280 | Q88HS3  | 3-oxoadipyl-CoA thiolase (EC 2.3.1.174) | Homology + synteny | Not in candidate list — ADD IT |
| paaI | PP_3281 | Q88HS2  | editing thioesterase                    | Homology + synteny | Accessory, not a core step     |

#### 4B. Peripheral / feeder genes (exclude from core chain)

- **peaE / PP\_3463** (phenylacetaldehyde dehydrogenase, EC 1.2.1.39): legitimate **upstream feeder** — oxidizes phenylacetaldehyde (from 2-phenylethylamine / styrene) to phenylacetate, the *substrate* of step 1. Flag as `feeder / peripheral`, not core.
- **phhA / PP\_4490** (Phe-4-hydroxylase, EC 1.14.16.1; GO:0006559 L-Phe catabolic process): routes Phe → Tyr; belongs to phenylalanine→tyrosine and downstream tyrosine catabolism. Peripheral.
- **hpd / PP\_3433** (4-HPP dioxygenase): tyrosine/homogentisate pathway (ppu00130). Not this module.
- **tyrB / PP\_1972, hisC / PP\_0967, amaC / PP\_3590** (aminotransferases, broad EC 2.6.1.-): broad transaminase over-propagation; not paa.

## 5. Gaps, Ambiguities, and Likely Over-Annotations

**Genuine metadata gap (genome-covered):** - **Step 7 thiolase paaJ / PP\_3280** absent from candidate list. Highest-priority fix. Genome status = covered.

**Paralog ambiguity (resolve by cluster membership + EC, never by EC alone):** - **Hydratase step (paaF, EC 4.2.1.17):** genome paralogs **PP\_2217** and **PP\_3726** are generic crotonase-superfamily enzymes (GO:0006635 fatty-acid  $\beta$ -oxidation), NOT cluster-resident. Do **not** assign them to the paa hydration step. `candidate_uncertain` if considered at all; correct assignment is PP\_3284. - **Thiolase step (paaJ, EC 2.3.1.174):** shared with **pcaF-I / PP\_1377** ( $\beta$ -ketoadipyl-CoA thiolase of the protocatechuate/ $\beta$ -ketoadipate pathway). Only **PP\_3280** belongs to paa.

**Likely over-annotation / noise in the candidate set (from broad ppu00360 map or shared EC):** - PP\_1791 (aldolase/synthase), PP\_2552 (DOPA decarboxylase), PP\_2932 (amidase), katG (PP\_3668, catalase-peroxidase = detox), hbd (PP\_3755), PP\_4983 (Trp 2-monooxygenase), dadA1/dadA2/PP\_4311 (D-amino-acid dehydrogenases, ppu00470). None participate in aerobic phenylacetyl-CoA catabolism.

**Confidence caveat:** every KT2440 paa protein is TrEMBL (unreviewed). EC numbers and subunit roles are automatic annotations; confidence rests on (i) tight cluster synteny, (ii) gene-for-gene homology to experimentally characterized *E. coli* W paa proteins, and (iii) direct KT2440 physiology for the activation step and operon regulation.

## 6. Module and GO-Curation Recommendations

**Per-step disposition:** - Steps 1, 2, 3, 4a, 4b, 5, 6 → **covered**. - Step 7 (thiolysis) → **covered** after adding PP\_3280/paaJ (currently would falsely read as **gap**). - paaI/PP\_3281 → **covered/accessory** (not a required reaction step).

**Module boundary:** - The generic module boundary is **correct** for this organism (canonical pathway, no lineage-specific rewiring). No `module_needs_revision` on biology grounds. - **However, the local bucket is mis-targeted:** `kegg:ppu00360` (broad Phenylalanine metabolism) is the reason ~15 spurious candidates appeared. **Recommend re-anchoring to the paa-specific module (KEGG M00878 / MetaCyc PWY-1341).** This is a metadata/bucket fix, not a biological revision.

**GO / new-document needs:** - No new GO *terms* need to be created — all activities have existing GO/EC coverage (e.g., GO:0010124 phenylacetate catabolic process; EC 1.14.13.149, 5.3.3.18, 3.3.2.12, 1.2.1.91). - **But GO annotation of the cluster is uneven and should be fixed.** A UniProt GO-BP scan shows only paaK, paaA, paaC, paaE, paaG, paaH, paaJ carry `phenylacetate catabolic process`, while **paaZ (PP\_3270) has no biological-process GO at all**, and **paaF (PP\_3284) is tagged only `fatty acid beta-oxidation`**; paaB (PP\_3277), paaD (PP\_3275), paaI (PP\_3281) also lack it. Request `GO:0010124 phenylacetate catabolic process` annotations for **paaZ, paaF, paaB, paaD, paaI**. - **Root cause of the dropped thiolase:** paaJ/PP\_3280 additionally carries an over-propagated `3,4-dihydroxybenzoate (protocatechuate) catabolic process` GO (from pcaF/PP\_1377 paralogy) — this cross-annotation is almost certainly why PP\_3280 was bucketed away from ppu00360 and omitted from the candidate list. Review/curate that GO on PP\_3280 to prevent recurrence.



**Not-expected-in-taxon:** none — the full pathway is expected and present.

## 7. Genes to Promote to Full fetch-gene Review

**High priority:** 1. **PP\_3280 / paaJ** — add to module as the step-7 thiolase; verify EC 2.3.1.174 and disambiguate from *pcaF* (PP\_1377). 2. **PP\_3270 / paaZ** — confirm bifunctional domain architecture (hydrolase EC 3.3.2.12 + ALDH EC 1.2.1.91) so steps 4a/4b are correctly attributed to one gene.

**Medium priority (paralog disambiguation):** 3. **PP\_3284 / paaF** vs paralogs **PP\_2217, PP\_3726** — lock the hydration step to the cluster copy. 4. **PP\_3275 / paaD** — obtain/confirm functional annotation (no EC in UniProt).

**Low priority (boundary confirmation):** *peaE*/PP\_3463 as a documented feeder; explicit exclusion of *phhA*, *hpd*, *tyrB*, *hisC*, *amaC*.

## 8. Key References

- **Ferrández et al. 1998**, *J Biol Chem* 273:25974–25986. PMID **9748275**. Defines the *E. coli* W paaZ / paaABCDEFGHIJK catabolon, PaaK ligase, PaaABCDE oxygenase, PaaZ third step, PaaFGHIJ  $\beta$ -oxidation-like segment, PaaX regulation. *Homology anchor for KT2440 (strong transfer: gene-for-gene syntenic cluster)*.
- **Kim et al. 2009**, *Environ Microbiol / J Bacteriol* work in KT2440. PMID **19406896**. *Direct target-strain evidence*: phenylacetyl-CoA ligase activity, PaaX-repressed *paa* operon, KDPG-mediated carbon catabolite repression of PAA metabolism.
- **Mohamed et al. 2002**, *J Bacteriol* 184:1953–1963. PMID **12189419**. *Azoarcus evansii* *paaYZGHKABCDE* cluster; PA activation to PA-CoA; broad conservation across bacteria. *Related-organism support (moderate transfer)*.
- **del Peso-Santos et al. 2008**, *J Bacteriol*. PMID **18544072**. *Pseudomonas* sp. Y2: PaaX/StyR regulation and *paaF*-named ligases; documents cross-organism gene-letter inconsistency (ligase called *paaF*). *Nomenclature caveat*.
- Pathway/module resources: KEGG map **ppu00360** (broad, current bucket), KEGG module **M00878** / MetaCyc **PWY-1341** (correct *paa*-specific target); UniProt proteome **UP000000556**.

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## Bottom line for curators

Mark the module **SATISFIED**. Make one substantive addition (**PP\_3280/paaJ** at step 7), retarget the bucket from the broad ppu00360 map to the paa-specific module, and prune the ~15 peripheral/over-propagated candidates. All KT2440 paa annotations are unreviewed (TrEMBL) but strongly supported by cluster synteny, *E. coli* homology, and direct KT2440 physiology for the activation/regulation steps.

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